keep it handy with yourself when you're actually programming the board and can look up something very easily.

Programming languages

The language you can use to write your program will vary from board to board. These section will list some of the major languages that you can use to write your programs.

Assembly

Assembly language is a low-level programming language. Instructions in assembly language also known as a mnemonic, usually maps one-to-one to the platform's machine code instructions which is collectively called it's instruction set. One of the drawback of assembly language is that the language is specific to an architecture, which is quite the opposite of portable high-level languages.

An assembly program is usually assembled by a program called an assembler. It essentially performs the act of converting assembly code to machine code. Assembly language is usually used in smaller projects which must have their softwares execute without a run-time components or libraries associated with a high-level languages. Examples of such systems are firmware for telephones, automobile fuel and ignition systems, air-conditioning control systems, security systems, and sensors. Programmers usually prefer high-level language like C to write their programs. Such languages abstract the details of the actual implementation from the programmers giving them a consistent and easy-to-use interface for writing programs. However some tasks can be performed better in assembly and some can performed only in assembly.

The following is an assembly program written for an Atmel AtMega16 microcontroller. It is written to solve the equation: c = x2 +2xy+y2

```
.include <m16def.inc>          ; description file for an AtMega16
.def temp = R17                ; alias for register R17
.def x = R18                   ; alias for register R18
.def y = R19                   ; alias for register R19

        LDI x, 0x0A            ; x = 10
        LDI y, C               ; y = 12
```